

EDUCATION

Philosophy of Doctorate, Informatics and Computing

Jan 2026– Dec 2029

United Arab Emirates University (UAEU), Abu Dhabi, UAE

- Supervisor: [Prof. Abdelkader Nasreddine Belkacem](#)
- Research Topic: “Decoding and Reconstructing Inner Speech Using EEG and ECoG for Neurorehabilitation”

Master of Science, Software Engineering

Oct 2018 – Dec 2022

University of Engineering and Technology (UET), Taxila, Pakistan

CGPA 3.33/4.00

- Supervisor: [Prof. Ali Javed](#)
- Thesis: “Deep Learning Model for the detection and classification of Diabetic Retinopathy”
- Major courses: Algorithm Design & Analysis, Advanced Software Technologies, Deep Learning

Bachelor of Science, Software Engineering

Oct 2014 – July 2018

Department of Software Engineering, University of Engineering & Technology (UET), Taxila, Pakistan

CGPA 3.50/4.00

- Supervisor: [Prof. Syed Muhammad Anwar](#)
- Final Year Project: “Context-Aware Smart Car App for Personal Assistance”
- Major courses: Artificial Intelligence, Web Development, Software Testing, Software Quality

RESEARCH INTERESTS

- Brain Computer interface (BCI) systems and neural signal processing using EEG and ECoG for cognitive and motor function analysis
- Decoding speech and inner speech from brain activity using deep learning for assistive communication and neurorehabilitation
- AI-driven neural decoding and simulation-based methods for brain-based communication and neurohabilitation applications

PUBLICATIONS

Toward Intelligent Breast Cancer Screening: Harnessing Capsule Networks for Robust Detection (IEEE-2026)
Samra Nawazish, Muhammad Zulkifl Hasan, Muhammad Amir Shahis, Noor Ul Hira, Nadeem Sarwar

Studying the Invasion of Drones in Indigenous areas using Machine Learning Techniques (Paper) (UCP Journal- 2024)
Samra Nawazish, Amin Ullah, Jawad Hassan

Performance Evaluation of MANET Routing Protocols Based on Size and Speed Parameters (Paper) (UCP Journal-2024)
Amin Ullah **Samra Nawazish**, Jawad Hassan

Image Fusion using Wavelet Transformation and XGboost Algorithm (Paper) (CMC-2024)
Faten S. Alamri, Dr Shahid Naseem, Tariq Mahmood, **Samra Nawazish**, Tanzila Saba

Towards sustainable smart cities: Applications, Challenges and Future Directions (Paper) (IJDSA-2025)
Amin Ullah, Saddam Hussain, **Samra Nawazish**, Irfan Ullah, Zafar Ali

Enhancing Dental Caries Identification with Deep Learning: A study of Convolutional Networks and Transfer Learning Approaches (Paper) (ssrn 4945802-2024)
Fayqa Manan, Muhammad Adeel, **Samra Nawazish**, Amin Ullah, Amjad Rehman

Deep Learning Model for the detection and classification of Diabetic Retinopathy (Paper) (ICTAPP-2023)
Samra Nawazish, Ali Javed, Tahira Nazir

PROFESSIONAL EXPERIENCE

Researcher

April 2025 – Dec 2025

[Intelligent Machines Lab](#), Information Technology University (ITU), Lahore, Pakistan

Advisor: [Dr. Waqas Sultani](#)

- Worked on real-time Microscopic Images and Video Enhancement and Stitching tool
- Trained [Segformer](#) on Hemato dataset to accurately generate segmentation masks for Blood Cancer dataset Leukemia
- Using [Transformers](#), developed a language translator that translates given text in English to Urdu
- Implemented [DRA-Net](#) for semantic cell segmentation on the BCCD dataset in Whole Slide Images
- Applied [ResNet-50](#) to classify flowers across seven categories

Teaching Assistant

Aug 2025 – Dec 2025

Instructor: [Dr. Mohsen Ali](#) Postgraduate Course Computer Vision CV525, ITU, Lahore, Pakistan

Graduate Research Assistant

Aug 2021 – Jan 2023

Department of Software Engineering, University of Engg. and Tech., Taxila, Pakistan

Research topic: Deep Neural Networks

- Explored Deep Learning models and transfer learning algorithms for the detection and classification of Diabetic Retinopathy

Lecturer

July 2021 – Dec 2024

Swedish College of Engineering & Technology, Wah Cantt, Pakistan

- Taught Courses: Programming Fundamentals, Object Oriented Programming, Introduction to Computers, Software Testing, Artificial Intelligence

Junior Software Developer

June 2019 – June 2021

Heavy Industries Taxila (HIT), Taxila Cantt, Pakistan

- I design, build, and maintain front-end, back-end, and database systems to deliver seamless, user-friendly applications

SUMMER SCHOOLS / INTERNSHIPS / WORKSHOPS / CONFERENCES

- Attended RISE-MICCAI Summer School (virtual) from July 14–18, 2025
- Attended RISE-MICCAI Summer School-Local Activity (virtual) from July 14-17, 2025 organized by MBZUAI
- Attended 2024 IEEE / OES China Ocean Acoustic Conference on May 29-31, 2024 in Harbin, China
- Attended China-Pakistan Marine Information Workshop and Training Course organized at Harbin Engineering University, Harbin, China from June 01-14, 2024
- Attended International conference on computing technologies, tools, and applications ICTAPP-23 on May 09-11, 2023, in Peshawar, Pakistan

HONOURS & AWARDS

- Fully funded Masters scholarship holder from UET Taxila, Pakistan
- Undergraduate merit-based scholarship holder from Federal Government Employee Quota
- Awarded Prime Minister Laptop Scheme Gift 2016
- Academic board topper excellence award in ICS (FBISE Islamabad) 2014
- Academic board topper excellence award in SSC (FBISE Islamabad) 2012

ACADEMIC SERVICES

- Served as a reviewer for MICCAI 2025
- Volunteer reviewer of Information fusion (in the area of medical imaging and IoMT)
- Designed Course Learning Outcomes and Program Learning Outcomes at Swedish College

LANGUAGES

English, Urdu, Punjabi

OTHER INTERESTS

- Exploring culture, Gardening, Cooking, Watching Tech videos

REFERENCES

- Dr. Ali Javed, Associate Professor, University of Engineering & Technology, Taxila Pakistan
Email: ali.javed@uettaxila.edu.pk Contact : +92334 5174516
- Dr. Amin Ullah, Associate Professor, Bahria University Lahore Campus, Lahore Pakistan
Email: aminullah.bulc@bahria.edu.pk Contact : +92333 9557815
- Dr. Tariq Mahmood, Post-doctorate Research Fellow, AIDA Lab, Prince Sultan University, Saudi Arabia
Email: tmahmood@psu.edu.sa Contact : +966532042334